

A nonatomic counterexample to the Measure Minimization Conjecture

Counterexample packet for arXiv:2308.00312

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Status. Full counterexample to Conjecture 3.2 as stated; likely valid, pending human review.

1 Source conjecture

Krishna's Measure Minimization Conjecture [1] asserts the following. Let $\{\tau_\alpha\}_{\alpha \in \Omega}$ be a continuous frame for a Hilbert space $\mathcal{H}$. If

$$h = \theta_\tau^* f, \quad f \in L^2(\Omega, \mu),$$

and

$$\mu(\text{supp } f) < \frac{1}{2} \left(1 + \frac{1}{\sup_{\alpha \neq \beta} |\langle \tau_\alpha, \tau_\beta \rangle|} \right), \quad (1)$$

then f is claimed to be the unique minimizer of support measure among all $g \in L^2(\Omega, \mu)$ satisfying $\theta_\tau^* g = h$.

Conjecture 3.2. (*Measure Minimization Conjecture*) *Let $\{\tau_\alpha\}_{\alpha \in \Omega}$ be a continuous frame for a Hilbert space $\mathcal{H}$. If $h \in \mathcal{H}$ can be written as*

$$h = \theta_\tau^* f \quad \text{for some } f \in \mathcal{L}^2(\Omega, \mu) \quad \text{with } \mu(\text{supp}(f)) < \frac{1}{2} \left(1 + \frac{1}{\sup_{\alpha, \beta \in \Omega, \alpha \neq \beta} |\langle \tau_\alpha, \tau_\beta \rangle|} \right),$$

then f is the unique solution of the measure minimization problem

$$(P_M) \quad \text{minimize } \{\mu(\text{supp}(g)) : g \in \mathcal{L}^2(\Omega, \mu)\} \quad \text{subject to } h = \theta_\tau^* g.$$

2 Proof intuition

Cardinality sparsity in a finite dictionary cannot be reduced continuously: one cannot use half an atom. Lebesgue support measure behaves differently. For a constant continuous frame, a coefficient with a prescribed integral can be compressed into an arbitrarily small interval simply by increasing its amplitude. The frame remains Parseval and the conjectured coherence threshold remains fixed, but the support-measure problem has infimum zero. The formal example below implements exactly this obstruction.

3 Counterexample

Theorem 1. *Conjecture 3.2 of [1] is false for a normalized continuous Parseval frame on the one-dimensional real Hilbert space.*

Proof. Let

$$(\Omega, \mu) = ([0, 1], \lambda), \quad \mathcal{H} = \mathbb{R}, \quad \tau_\alpha = 1 \quad (\alpha \in [0, 1]).$$

This is a normalized continuous Parseval frame: for every $h \in \mathbb{R}$,

$$\int_0^1 |\langle h, \tau_\alpha \rangle|^2 d\alpha = \int_0^1 |h|^2 d\alpha = |h|^2.$$

Its synthesis operator is

$$\theta_\tau^* f = \int_0^1 f(\alpha) \tau_\alpha d\alpha = \int_0^1 f(\alpha) d\alpha,$$

and its off-diagonal coherence is

$$\sup_{\alpha \neq \beta} |\langle \tau_\alpha, \tau_\beta \rangle| = 1.$$

Fix any $a \in (0, 1)$ and set

$$f_a = \frac{1}{a} \mathbf{1}_{(0, a)} \in L^2([0, 1]).$$

Then

$$\theta_\tau^* f_a = \frac{1}{a} \lambda(0, a) = 1, \quad \mu(\text{supp } f_a) = a < 1 = \frac{1}{2} \left(1 + \frac{1}{1} \right).$$

Thus f_a satisfies every hypothesis of the conjecture for $h = 1$.

Now choose any $b \in (0, a)$ and define $f_b = b^{-1} \mathbf{1}_{(0, b)}$. Again $f_b \in L^2([0, 1])$ and

$$\theta_\tau^* f_b = 1 = \theta_\tau^* f_a, \quad \mu(\text{supp } f_b) = b < a = \mu(\text{supp } f_a).$$

Therefore f_a is not even a minimizer of the problem (P_M) , and hence cannot be its unique minimizer. Since b can tend to zero, the infimum of the support measure in the synthesis fiber of $h = 1$ is zero; it is not attained because the zero-support equivalence class synthesizes zero. $\square$

4 Verification, scope, and novelty

The accompanying exact-arithmetic script checks the frame identity, coherence, threshold, synthesis values, and strict support decrease for several rational pairs $0 < b < a < 1$. These checks are illustrative; the displayed calculation is the proof.

The counterexample resolves only Conjecture 3.2. It does not resolve the separate questions about equality, endpoint versions, and Tao analogues, or the NP-hardness classification in Problem 3.3. The obstruction suggests that a repaired statement must impose atomicity or amplitude control, or replace support measure by a coercive objective such as total variation.

A bounded search on 11 August 2026 used the exact conjecture name and formula, the source id and author, and continuous-frame support-minimization phrases. Later related preprints by the same author on continuous Deutsch, spark, and Welch inequalities were checked at title/abstract level. No prior counterexample or retraction was located. Novelty confidence is moderate because the example is elementary.

Human-review recommendation. Verify the standard definition of a continuous Parseval frame and the one-line synthesis calculation. If confirmed, classify this as a full counterexample to Conjecture 3.2 as literally stated.

References

- [1] K. M. Krishna, *Functional Continuous Uncertainty Principle*, arXiv:2308.00312 (2023), Conjecture 3.2.